Q.27 Explain programming terminal for PLC. (CO3)

Q.28 Explain comparison instruction of PLC like equal, not equal, greater, greater than equal to. (CO3)

Q.29 Enlist five advantages of PLC in industry. (CO5)

Q.30 Explain the different programming languages of PLC. (CO4)

Q.31 Discuss importance of Local area Network for DCS. (CO3)

Q.32 Explain I/O structure of PLC. (CO3)

Q.33 What do you mean by SCADA? What are its applications? (CO6)

Q.34 Explain the methods of speed control of motor. (CO7)

Q.35 Difference between open architecture and dedicated system. (CO6)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 What is automation? Explain generalized automation, production systems and their classification. (CO1)

Q.37 Draw block diagram of PLC and explain function of each block in detail. (CO1)

Q.38 What are AC drives? Explain different types of AC drives. (CO3)

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6th Sem / Eltx

Sub : Industrial Automation

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 The PLC is used in ______. (CO2)

a) Machine tools

b) Automated assembly equipment

c) Molding and extrusion machines

d) All of the above

Q.2 The PLCs were originally designed to replace (CO1)

a) Analog controllers b) DCS

c) Microcomputers d) Hardwired control

Q.3 The ____________ is moved toward the relay electromagnet when the relay is on. (CO7)

a) Armature b) Coil

c) No contact d) NC contact

Q.4 What is the full form of SCADA? (CO7)

a) Supervisory Control & Document Acquisition

b) Supervisory Control & Data acquisition

c) Supervisory Column & Data Assessment

d) Supervisory Column & Data Assessment

Q.5 The control in SCADA is ______. (CO6)

a) Online control b) Direct control

c) Supervisory control d) Automatic control

Q.6 DCS is a _______. (CO7)

a) Distributed Control System

b) Data Control System

c) Data Column System

d) Distributed Column System

Q.7 The control in SCADA is _____. (CO7)

a) Online control b) Direct control

c) Supervisory control d) Automatic control

Q.8 What is the main function of the CPU in a PLC? (CO7)

a) To provide power to the PLC

b) To store the program

c) To execute the program and control the input and output devices

d) To communicate with other PLCs

Q.9 Solenoids, lamps, motors are connected to: (CO4)

a) Analog output b) Digital output

c) Analog input d) Digital input

Q.10 ______ of PLCs can be done in very little time. (CO2)

a) Programming b) Installation

c) Commissioning d) All of the above

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 There are _____ type of timers in PLC. (CO3)

Q.12 What is retentive timer? (CO2)

Q.13 PLC stands for ______. (CO1)

Q.14 _____ is a electromagnetic switch. (CO2)

Q.15 Expand EPROM. (CO1)

Q.16 Small PLCs have a memory from ______ to store the user's logic program. (CO2)

Q.17 What is counter? (CO1)

Q.18 _____ is the heart of the SCADA system. (CO6)

Q.19 A relay is an example of input device? (True/ False). (CO1)

Q.20 What is bus? (CO2)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain the operation of a PLC. (CO2)

Q.22 Explain real time clock function. (CO3)

Q.23 Write short note on memory structure of PLC. (CO2)

Q.24 Explain timer and counter instruction of PLC. (CO2)

Q.25 Explain roll of common system components of SCADA. (CO6)

Q.26 What are the applications of PLC in industry? (CO5)